

Selective Quad-Band Cross-Polarization and Multi-Band Linear-to-Circular Conversion Using an Asymmetric Single-Layer Reflective Metasurface

Abstract—A compact asymmetric reflective metasurface is presented for selective multi-band polarization control across the 5–20 GHz microwave range. The proposed unit cell operates with a single copper resonator and a continuous ground plane on a low-loss Rogers RT5880 substrate with an 8 mm periodicity and 1.8 mm dielectric thickness. Full-wave simulations show four distinct linear-to-cross-linear polarization-conversion (CPC) windows at 7.1–7.5, 9.7–9.9, 14.2–14.6, and 17.5–18.5 GHz, with near-unity conversion at 7.3, 9.8, 14.4, and 18.0 GHz. The same passive surface also generates intermediate linear-to-circular (LTC) states at 7.7, 9.4, and 10.0 GHz, corresponding to LHCP, LHCP, and RHCP, respectively. The CPC response remains highly efficient for incidence angles from 0° to 60° , demonstrating strong angular robustness. Compared with representative reported converters, the proposed structure combines four narrow CPC windows, three LTC windows, a low-loss single-layer realization, and 60° angular operation within one 5–20 GHz platform. The proposed architecture is therefore promising for polarization-selective microwave links, radar front ends, and satellite communication systems.

Index Terms—Metasurface, polarization converter, cross-polarization conversion, linear-to-circular conversion, polarization conversion ratio, angular stability, Rogers RT5880.

I. INTRODUCTION

The degree of polarization is a key parameter in microwave systems as it can dramatically influence antenna compatibility, channel utilization, radar signatures, and sensing performance. This has made metasurface-based converters attractive alternatives to traditional polarization control devices in the form of a planar converter. A compact asymmetric reflective metasurface has been demonstrated for simultaneous linear-to-cross-linear and linear-to-circular conversions with angular stability of 60° [1]. Anisotropic metasurfaces have also been used to demonstrate multi-band linear polarization conversion, establishing the ability to obtain multiple resonant responses in a small unit cell [2]. Tunable cross-polarization conversion using a graphene-based hollow-H structure has further demonstrated the possibility of dynamically controlled polarization manipulation [3]. A multiband plasmonic-resonance metasurface reported cross-polarization conversion together with linear-to-circular and circular-to-linear transformations over a broad frequency range and angular range [4]. Recently, coding metasurfaces have also been integrated with dual polarization conversion for wide bandwidth radar cross section reduction [5]. A low-profile broadband polarization-conversion metasurface has been combined with an antenna for RCS reduction with a wide angular operation [6]. Re-

cently, a multi-band ultrathin reflective metasurface has been designed for linear and circular polarization conversion in Ku, K, and Ka bands [7]. H-shaped anisotropic resonators have achieved ultra-broadband reflective linear polarization conversion, which can be made wide-angle with the help of multiple resonances [8]. An anisotropic metasurface has also been applied to achieve ultra-broadband linear polarization conversion in a broad microwave frequency spectrum [9]. Metasurfaces have also been designed to achieve highly efficient reflection-based linear-to-circular polarization conversion across a broad frequency range [10]. It has been reported that a single-layer low-cost metasurface can achieve linear-to-linear and linear-to-circular conversion in a few C-, X-, and Ku-band frequency bands [11]. An ultrathin reflective converter has also been integrated with LP-LP and LP-CP operation with angular stability for K- and Ka-band applications [12]. Multi-band reflective polarization conversion has also been explored for satellite-oriented applications, while maintaining useful conversion efficiency over several separated frequency regions [13]. An interesting multifunctional metasurface has been recently studied for the simultaneous linear-to-linear and linear-to-circular conversion as a single polarization-control function [14]. Several different conversion regions have been reported for the multifunctional wideband conversion of low-RCS radiating systems [15]. The transmission mode of dual-band dual-linear-to-circular conversion has been studied for satellite communication applications [16]. Low-loss broadband linear-to-circular conversion has also been achieved for K-/Ka-band satellite links [17]. Linear-to-circular and linear-to-cross-polarization converters (LTC and CPC) with angular stability have been proposed in Ka-band applications and are highly miniaturized [18]. A single-layered structure has also been shown to achieve dual-band linear-to-dual-circular conversion, which is enhanced by angle stability [19]. Ultrathin reflective linear and circular polarization conversion has also been studied for Ku-band operation [20].

Even with these advances, there is always a compromise between the number of windows for conversion, the efficiency of this conversion, angular stability, profile, and structural simplicity. Many reported structures highlight single- (or two-) wide conversion bands, while a minority of designs explicitly add some separated cross-polarization windows with intermediate circular-polarization windows in a single passive layer. This encourages a design using *selective multifunctionality* (as opposed to a single, broadband response).

This paper proposes an asymmetric single-layer reflective metasurface which can be used for selective polarization conversion in the frequency range 5–20 GHz. The proposed structure has four high-efficiency CPC windows and three intermediate LTC states in one passive unit cell. Additionally, the response of the CPC is evaluated for incidence angles of 0° to 60° . The key contributions are the selective multifunctional conversion states, the low-loss single-layer configuration, and the stable operation over a wide range of incidence angles within a compact reflective platform.

II. UNIT-CELL DESIGN AND METHODOLOGY

A. Geometry and Materials

The unit cell is periodic with $p = 8.0$ mm and it is composed of an asymmetric square resonator with a circular slot and a narrow cutting slot. The principal dimensions are $a = 6.0$ mm, $s = 4.825$ mm, $R = 1.4$ mm, and $c_w = 0.2$ mm. The metallic layer is copper and the thickness of the metallic layer is $t = 0.035$ mm, the dielectric substrate is Rogers RT5880 and the thickness is $h = 1.8$ mm, the relative permittivity is about 2.2, the loss tangent is about 0.0009. A continuous metallic ground plane is placed beneath the dielectric substrate. The broken rotational symmetry results in differing electromagnetic responses along orthogonal axes, which is needed for polarization conversion.

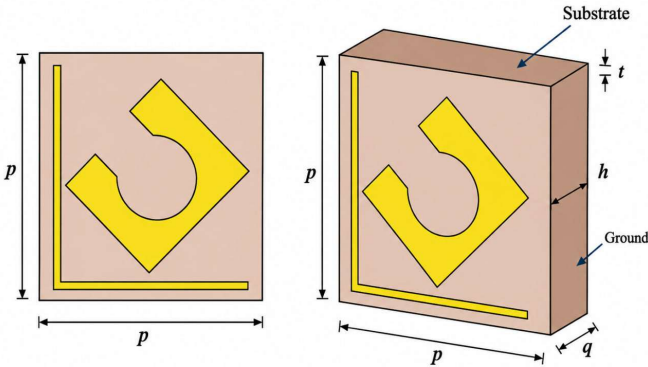


Fig. 1: Front and perspective views of the proposed unit cell.

TABLE I: Optimized geometric parameters of the proposed unit cell.

Parameter	Description	Value (mm)
p	Unit-cell periodicity	8.000
a	Main resonator side length	6.000
s	Inner-to-outer resonator distance	4.825
R	Circular-slot radius	1.400
c_w	Cutting-slot width	0.200
h	RT5880 thickness	1.800
t	Copper thickness	0.035

B. Electromagnetic Formulation

For a linearly polarized incident wave, the reflected field is described through the Jones reflection matrix

$$\mathbf{R} = \begin{bmatrix} R_{xx} & R_{xy} \\ R_{yx} & R_{yy} \end{bmatrix}. \quad (1)$$

The eigenvalue relation $\mathbf{R}\mathbf{X} = m\mathbf{X}$ is used to interpret the principal polarization axes. For y -polarized incidence, the polarization conversion ratio is calculated as follows:

$$\text{PCR} = \frac{|R_{xy}|^2}{|R_{xy}|^2 + |R_{yy}|^2}. \quad (2)$$

The normalized ellipticity is used to identify the handedness of the reflected wave. In the adopted convention, $e = 0$ corresponds to linear polarization, $e = -1$ to LHCP, and $e = +1$ to RHCP. An extra circular-polarization indicator, the axial ratio (AR), is used. Ideal circular polarization is achieved when AR is close to 0 dB. The complex co- and cross-polarized reflection coefficients are exported from CST and numerically post-processed to obtain these quantities.

C. Simulation Procedure

The structure is modeled in CST Microwave Studio with periodic boundary conditions in the transverse directions and Floquet-port excitation in the propagation direction. A frequency sweep from 5 to 20 GHz is used. The study is done at normal incidence for both x - and y -incidence, and at angles of 0° – 60° for the angular study. The optimizations focus on high CPC efficiency at main resonances, on useful LTC states between the CPC bands, on a low profile, and on stable response to oblique incidence.

III. RESULTS AND DISCUSSION

A. Reflection Characteristics

The reflection coefficients in linear scale are shown in Fig. 2. For orthogonal incident polarizations, the co-polarized terms have noticeable dips and the cross-polarized terms have peaks at the same frequencies. Four principal resonances occur near 7.3, 9.8, 14.4, and 18.0 GHz. The reflected cross-polarized magnitudes are close to 0.95–0.98, indicating efficient conversion of the incident linear polarization into its orthogonal reflected state.

The dB-scale response in Fig. 3 further confirms the inverse relation between co- and cross-polarized components. The co-polarized minima are approximately -26 dB, -30 dB, -16 dB, and -31 dB at 7.3, 9.8, 14.4, and 18.0 GHz, respectively. At these points, the cross-polarized response is closer to 0 dB. Therefore, the resonant energy is mainly transferred to the orthogonal polarization channel instead of the co-polarized channel.

B. CPC and Circular-Polarization Behavior

The PCR curves are plotted in Fig. 4. The PCR approaches unity at the four main resonances, defining selective CPC windows of 7.1–7.5, 9.7–9.9, 14.2–14.6, and 17.5–18.5 GHz. The combined bandwidth of these four CPC windows is

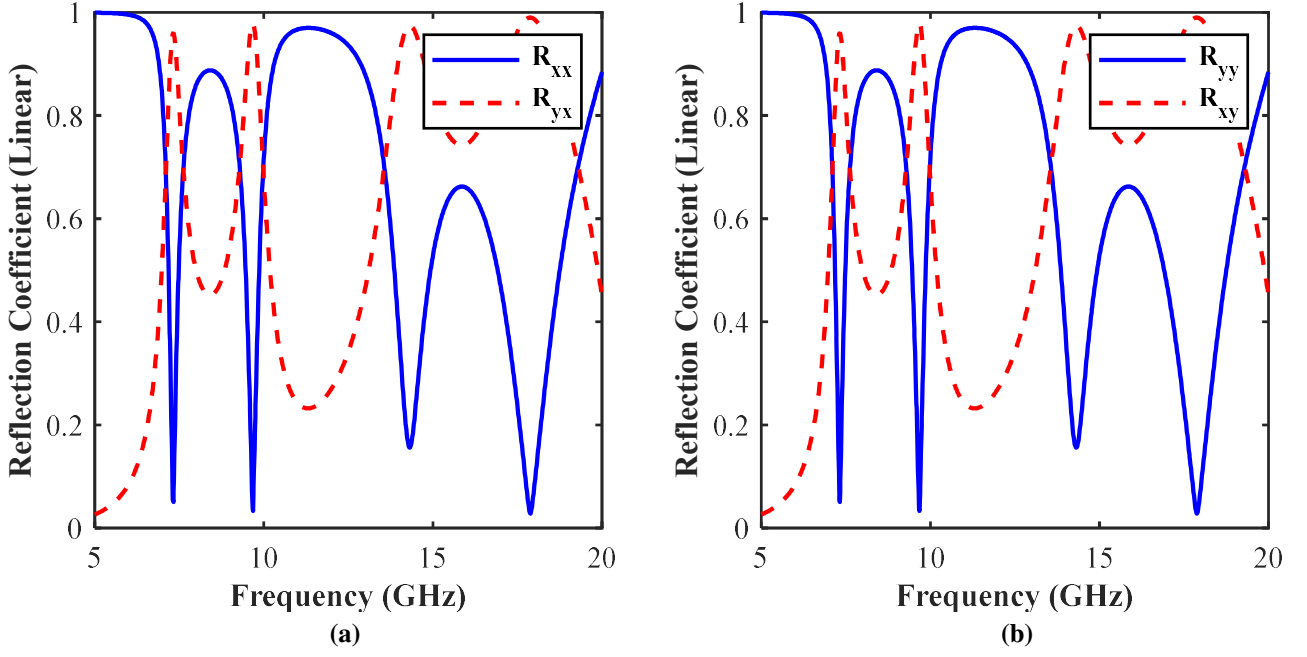


Fig. 2: Linear-scale co- and cross-polarized reflection coefficients for (a) x -polarized and (b) y -polarized incidence.

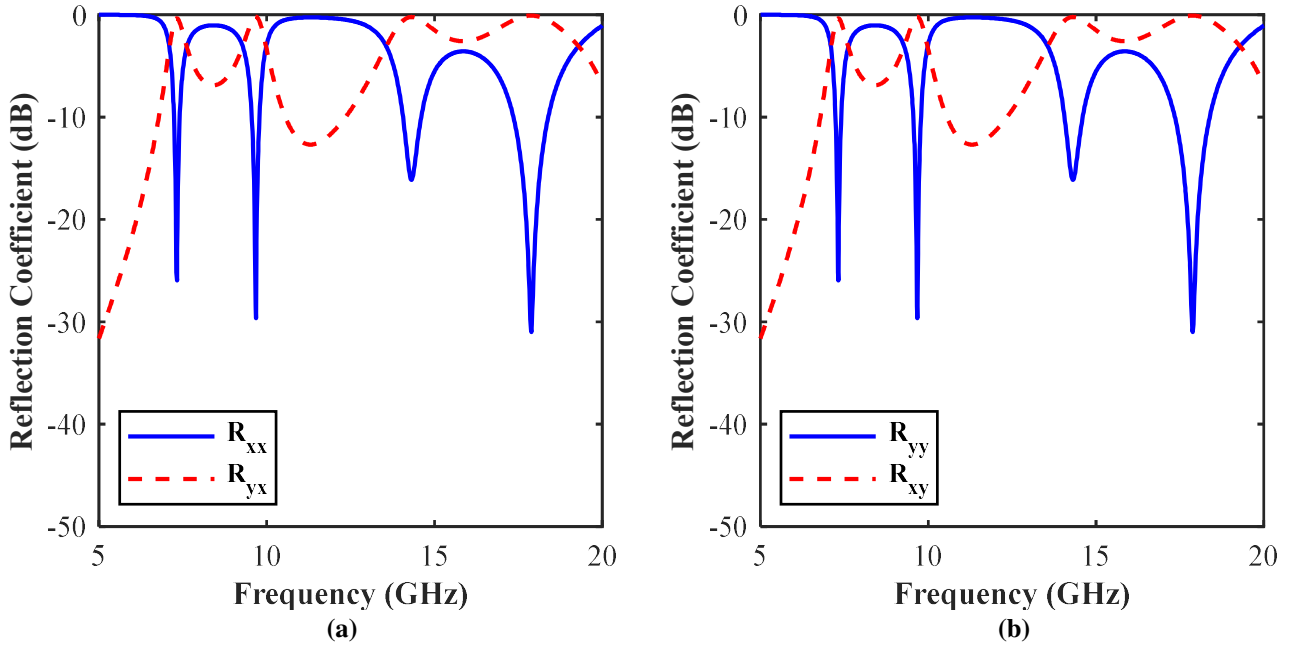


Fig. 3: dB-scale co- and cross-polarized reflection coefficients for (a) x -polarized and (b) y -polarized incidence.

approximately 2.0 GHz over the 5–20 GHz range. The high PCR values confirm efficient conversion into the orthogonal reflected polarization, while the separated-band response enables frequency-selective polarization conversion at designated microwave channels.

The ellipticity response is presented separately in Fig. 5. It approaches zero near the main CPC resonances, indicating a linearly polarized cross-converted reflected field. Between

these resonances, the ellipticity approaches -1 near 7.7 and 9.4 GHz and approximately $+1$ near 10.0 GHz, corresponding to LHCP, LHCP, and RHCP, respectively. These responses identify three LTC windows at 7.6–7.8, 9.3–9.5, and 9.9–10.1 GHz. Overall, the proposed metasurface provides seven distinct polarization-conversion windows across the 5–20 GHz range, comprising four CPC and three LTC bands.

The PCR and ellipticity responses identify four CPC and

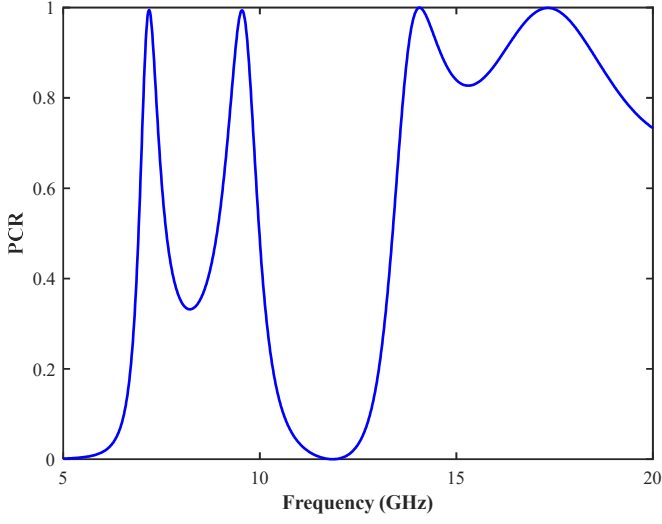


Fig. 4: PCR with normal incident wave.

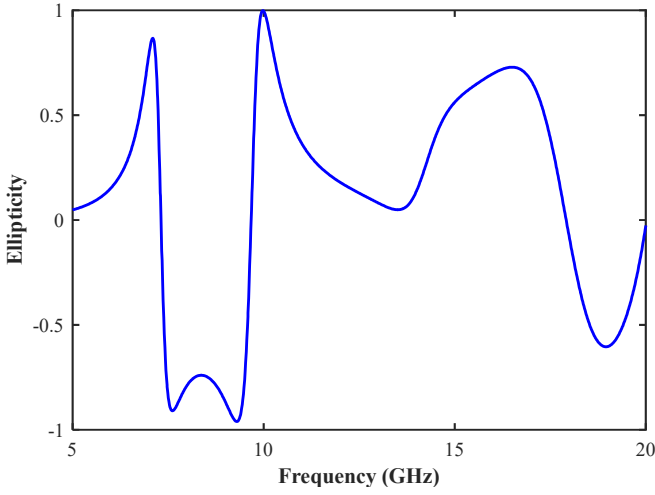


Fig. 5: Ellipticity with normal incident wave.

three LTC bands within the 5–20 GHz range. The CPC windows occur at 7.1–7.5, 9.7–9.9, 14.2–14.6, and 17.5–18.5 GHz, while the LTC windows occur at 7.6–7.8, 9.3–9.5, and 9.9–10.1 GHz. As summarized in Table II, these seven windows demonstrate the frequency-selective polarization-conversion capability of the proposed metasurface.

TABLE II: Principal polarization-conversion windows extracted from the simulated response.

Mode	Frequency (GHz)	Center (GHz)
CPC-1	7.1–7.5	7.3
LTC-1	7.6–7.8	7.7
LTC-2	9.3–9.5	9.4
CPC-2	9.7–9.9	9.8
LTC-3	9.9–10.1	10.0
CPC-3	14.2–14.6	14.4
CPC-4	17.5–18.5	18.0

Further evidence of the circular-polarization characteristics

of the proposed metasurface can be found in the AR response shown in Fig. 6. There are several sharp minima in the frequency regions where the ellipticity suggests that the state is circular or near-circular. The relative amplitude and phase of the orthogonal reflected components form the basis of AR and complement the ellipticity analysis. Together, the PCR, ellipticity, and AR responses distinguish the major CPC resonances from the intermediate circular-polarization states.

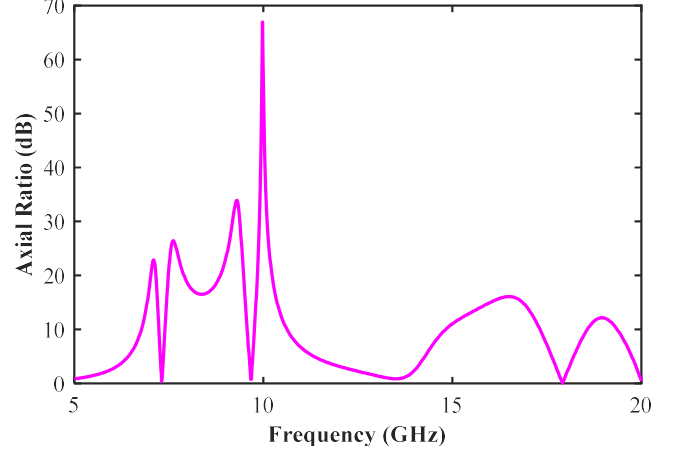


Fig. 6: Axial ratio of the reflected wave over 5–20 GHz.

C. Angular Stability

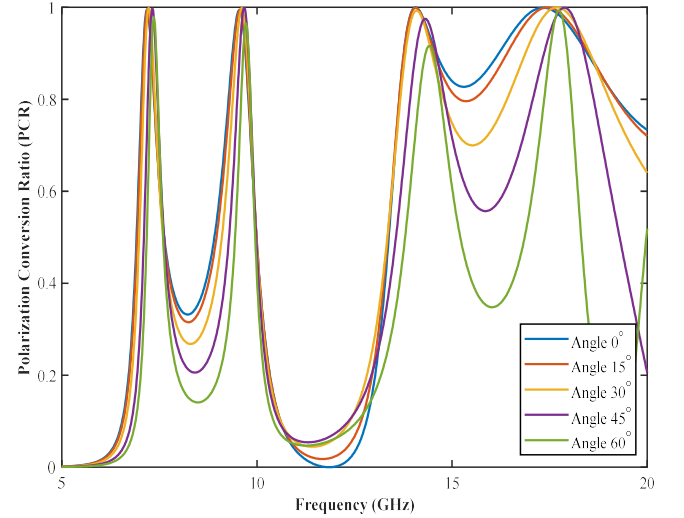


Fig. 7: PCR response at incidence angles of 0°, 15°, 30°, 45°, and 60°.

The angular stability of the proposed metasurface is evaluated for incidence angles of 0°, 15°, 30°, 45°, and 60°, with the corresponding PCR responses shown in Fig. 7. The four main resonant features remain discernible up to 60°, indicating strong angular stability. The principal PCR peaks remain nearly unity at 0°, 15°, and 30°, with only small spectral shifts. At 45° and 60°, the resonances become more selective, while

TABLE III: Comparison with representative reported polarization converters.

Reference	Main CPC/LP-LP operation	LTC/LP-CP operation	Angular range	Key distinction
[1]	15.5–16.5 GHz	13 and 18 GHz CP states	60°	Asymmetric multifunctional reflective converter
[4]	Five CPC bands	Eight LP/CP transformation bands	60°	Plasmonic multiband multifunctionality
[11]	6.46–6.78; 10.52–11.85; 16.49–17.37 GHz	7.28–9.40; 13.38–15.19 GHz	45°	Low-cost single-layer LP/CP converter
[12]	18.55–26.90; 43.55–45.30 GHz	31.8–40.5 GHz	40°	1-mm ultra-thin K/Ka-band converter
[13]	Multi-band linear conversion	Multi-band circular conversion	–	Reflective multi-band polarization conversion
This Work	7.1–7.5; 9.7–9.9; 14.2–14.6; 17.5–18.5 GHz	7.6–7.8; 9.3–9.5; 9.9–10.1 GHz	60°	Four CPC + three LTC windows; RT5880; single layer

good conversion is retained at the principal bands. Overall, the results show only a gradual angular dependence without significant degradation of the main conversion response.

IV. PHYSICAL INTERPRETATION AND COMPARATIVE ASSESSMENT

The asymmetric resonator has an asymmetric unit cell that results in different complex reflection responses along principal axes. The incident field can be decomposed into two orthogonal components in the eigen-polarization picture, whose amplitudes and phases are manipulated by the resonant surface. If the reflected components have the proper phase relationship to produce orthogonal linear polarization, a high PCR will be achieved. However, when the two components have almost equal amplitude and a quadrature phase difference, the reflected field is almost circularly polarized. Multiple resonant conditions are introduced with the combination of the main patch, circular slot, cutting slot, and relative spacing of the slots, without having to introduce many layers into the configuration. It is observed that this is the typical behavior of the four important PCR maxima and three intermediate ellipticity extrema, obtained in the results.

Another feature of the proposed design is its frequency-selective characteristics. The metasurface generates multiple discrete conversion bands rather than a single continuous conversion bandwidth. (Table II). Separated bands can be advantageous for frequency-selective polarization control, for which different microwave channels might need different polarization states.

Table III compares the proposed converter with the available designs. There are several reported structures that offer wide LP-LP conversion, more than one circular-polarization band, or good angular stability. The four selective CPC bands and three intermediate LTC states in the 5–20 GHz frequency range, however, provide a unique multifunctional response. The proposed structure also preserves its primary characteristics of conversion up to incidence angles of 60°. Furthermore, the 8 mm periodicity and 1.8 mm substrate thickness contribute to the compact electrical size, resonant selectivity, and low dielectric loss of the structure.

Instead of trying to achieve a single continuous conversion bandwidth, the unit cell is used to obtain separate conversion windows at particular frequencies. This approach enables different polarization states to be obtained from the same passive surface at different frequency channels. In addition, the overall structure is greatly simplified using the single-layer configuration, while maintaining the desired multiband polarization-conversion characteristics.

V. CONCLUSION

A selective multi-band reflective polarization converter has been analyzed over 5–20 GHz using an asymmetric single-layer unit cell on Rogers RT5880. The structure provides four high-efficiency CPC bands centered at 7.3, 9.8, 14.4, and 18.0 GHz, with near-unity PCR, while ellipticity analysis identifies three intermediate LTC states around 7.7, 9.4, and 10.0 GHz. The principal CPC response remains strong under incidence angles up to 60°, although the highest-angle response exhibits increased spectral selectivity and resonance displacement. The resulting combination of seven conversion windows, low-loss substrate, single-layer construction, and wide-angle operation makes the proposed architecture a promising candidate for polarization-selective radar, satellite, and microwave communication front ends. Future work should experimentally validate the structure and quantify fabrication tolerance and finite-array effects.

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